

Java Comments

Documenting logic for humans, ignored by machines.

What Are Comments?

Comments are programmer notes that tell Java to "skip" specific parts of the file. They serve as a roadmap for anyone reading your code, including your future self.

1. Single-Line Labels

Starts with `//`. Everything to the right on that specific line is ignored.

```
// This is a note  
int x = 5;
```

2. Multi-Line Blocks

Starts with `/*` and ends with `*/`. Useful for long explanations or headers.

```
/* This block spans  
multiple lines */
```

3. Disabling Code

Comments are vital for **debugging**. If a piece of code is causing errors, you can "comment it out" instead of deleting it to see if the program runs without it.

```
System.out.println("Running ... ");  
// System.out.println("Wait, I might break things!");
```

4. Best Practices: "The Why, Not the What"

Junior developers often comment things that are already obvious. Professional comments explain the logic behind a decision.

✗ AVOID (Over-commenting)

```
int age = 18; // store age
```

The code already says it stores age. This is clutter.

✓ PREFER (Explain Reason)

```
int retryLimit = 3; // API security  
limit
```

This tells us the purpose or the limit's source.

5. Nested Comment Warning

Unlike some other languages, Java does **not** allow you to place a multi-line comment inside another multi-line comment. This will confuse the compiler and cause an error.



INVALID SYNTAX:

```
/* Outer comment /* Nested comment */ */
```

The first `*/` Java sees will end the entire comment block.

6. Real-World Implementation

Here is how a clean, professional Java file looks with properly utilized comments:

```
public class ScoreManager {
    public static void main(String[] args) {
        // Initialize starting points
        int score = 50;

        /* Apply bonus multiplier
           based on holiday event logic */
        score = score * 2;

        System.out.println("Total: " + score); // Display final result
    }
}
```



Key Takeaways

- **Ignored by JVM:** Comments have zero impact on speed.
- **Single-Line:** Uses `//` for quick notes.
- **Multi-Line:** Uses `/* */` for blocks.
- **Better Debugging:** Comment out code to isolate bugs.

- **Explain "Why":** Focus on logic, not syntax.
- **Cleanliness:** Avoid commenting on obvious code.
- **Header notes:** Use blocks for file descriptions.
- **No Nesting:** Don't put multi-line inside multi-line.

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